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CASE STUDY

Orthonormal Compactly Supported Wavelets with Optimal Sobolev Regularity: Numerical Results

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Abstract—Numerical optimization is used to construct new orthonormal compactly supported wavelets with a Sobolev regularity exponent as high as possible among those mother wavelets with a fixed support length and a fixed number of vanishing moments. The increased regularity is obtained by optimizing the locations of the roots the scaling filter has on the interval $(\pi/2, \pi)$. The results improve those obtained by I. Daubechies (1988, *Comm. Pure Appl. Math.* **41**, 909–996), H. Volkmer (1995, *SIAM J. Math. Anal.* **26**, 1075–1087), and P. G. Lemarié-Rieusset and E. Zahrouni (1998, *Appl. Comput. Harmon. Anal.* **5**, 92–105).

Key Words: wavelet; regularity; Sobolev exponent.

1. INTRODUCTION

To construct orthonormal compactly supported wavelets we use as the starting point the dilation equation

$$\phi(x) = \sqrt{2} \sum_{k \in \mathbb{Z}} c_k \phi(2x - k), \qquad x \in \mathbb{R},\ c_k \in \mathbb{R}, \tag{1}$$

for the scaling function or father function ϕ. A solution exists in the sense of distributions when $\sum_k c_k = \sqrt{2}$ and it has compact support if only finitely many of the filter coefficients c_k are nonzero.

The condition when the solution to (1) yields wavelets can be expressed in terms of the scaling filter

$$m_0(\xi) = \frac{1}{\sqrt{2}} \sum_{k \in \mathbb{Z}} c_k e^{-ik\xi}. \tag{2}$$

If m_0 satisfies

$$|m_0(\xi)|^2 + |m_0(\pi + \xi)|^2 = 1, \qquad \xi \in \mathbb{R}, \tag{3}$$

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$m_0(0) = 1$, and a technical condition called the Cohen criterion holds, then there exists a scaling function and wavelet pair [1, 3, Section 6.3]. We will not need exact details about Cohen's condition; we only use that if m_0 does not vanish on $[-\pi/2, \pi/2]$, then it satisfies the criterion.

If ϕ is such that it gives rise to a wavelet, the wavelet ψ is given by

$$\psi(x) = \sum_{k\in\mathbb{Z}} (-1)^k c_{1-k}\phi(2x-k).$$

Note that then ϕ and ψ have the same regularity properties.

The original construction of orthonormal compactly supported wavelets by Daubechies [2] considers a trigonometric polynomial m_0 with at most $2N$ nonzero coefficients $c_0, \ldots, c_{2N-1}$. The polynomial m_0 is required to satisfy (3) and to have a zero at π of maximal order N (the order of the zero is also the number of vanishing moments the corresponding wavelet has). The resulting equations can be solved and the constructed m_0 has no roots on $(-\pi, \pi)$; hence it satisfies the Cohen criterion. (Ref. [3] is a comprehensive reference for these results.)

To construct smoother wavelets our approach is to reduce the order of the zero m_0 has at π and instead to introduce zeros on the interval $(\pi/2, \pi)$. Keeping N and the number of zeros on $(\pi/2, \pi)$ fixed we use numerical optimization to choose the locations of the roots so that the resulting wavelets are as smooth as possible. We are able to construct wavelets that are more regular when compared with those introduced in [2, 5, 9] with the same number of filter coefficients.

A detailed description of the methods used and analysis of the results is presented in [7]. Sample Mathematica programs, the results of the optimizations, and a complete listing of filters for the most regular wavelets are available from the author.

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2. RESULTS

In the numerical optimization for smoothness we studied wavelets with up to 40 filter coefficients and whose scaling filter has up to four roots on the interval $(\pi/2, \pi)$. The Sobolev regularity exponents of the most regular wavelets found are shown in Table I. By the Sobolev regularity exponent s_0 of ϕ we mean $s_0 = \sup\{s : \phi \in H^s\}$, where H^s is the Sobolev space consisting of all $f \in L^2(\mathbb{R})$ such that $(1+|\xi|^2)^{s/2}\hat{f}(\xi) \in L^2(\mathbb{R})$. The length of the filter m_0 is $2N$ and the filter has n_z roots on the interval $(\pi/2, \pi)$. The results are also summarized in Fig. 1.

The column $n_z = 0$ gives the Sobolev exponents for the Daubechies wavelets constructed in [2]. For $N \geq 4$ we have constructed wavelets which are more regular. For large values of N there are four families of wavelets corresponding to different numbers of roots on $(\pi/2, \pi)$, and the regularity increases with the number of roots.

To relate these results to Hölder regularity, it can be shown that the Hölder regularity exponent $\alpha_0 = \sup\{\alpha : \phi \in C^\alpha\}$ satisfies $\alpha_0 \in [s_0 - 1/2, s_0]$ [8, Corollary 9.9]. Here C^α consists of functions that are continuously differentiable up to order equal to the integer part of α, and the highest order continuous derivative belongs to the (standard) Hölder class with exponent equal to the fractional part of α.

TABLE I
The Best Sobolev Regularity Exponents Found

	Sobolev regularity exponent s_0				
$2N$	$n_z = 0$	$n_z = 1$	$n_z = 2$	$n_z = 3$	$n_z = 4$
2	0.50				
4	1.00				
6	1.42	1.00			
8	1.78	1.82			
10	2.10	2.26	1.00		
12	2.39	2.66	2.00		
14	2.66	3.02	3.00	1.00	
16	2.91	3.37	3.48	2.00	
18	3.16	3.72	3.92	3.00	1.00
20	3.40	4.07	4.32	4.00	2.00
22	3.64	4.42	4.73	4.76	3.00
24	3.87	4.78	5.14	5.22	4.00
26	4.11	5.14	5.55	5.67	5.00
28	4.34	5.50	5.95	6.10	5.84
30	4.57	5.85	6.34	6.51	6.38
32	4.79	6.19	6.71	6.89	6.88
34	5.02	6.52	7.09	7.25	7.30
36	5.24	6.83	7.45	7.62	7.69
38	5.47	7.15	7.81	7.99	8.08
40	5.69	7.46	8.17	8.37	8.51

Note that in each column $n_z = 1, \ldots, 4$, the first few entries are smaller than in the previous columns on the same row. This is not surprising, since it can be shown that in order to have $\psi \in H^n$, the wavelet ψ must have at least $n + 1$ vanishing moments [8, Proposition 8.3]. The wavelets corresponding to the entries in the table have $N - 2n_z$ vanishing moments.

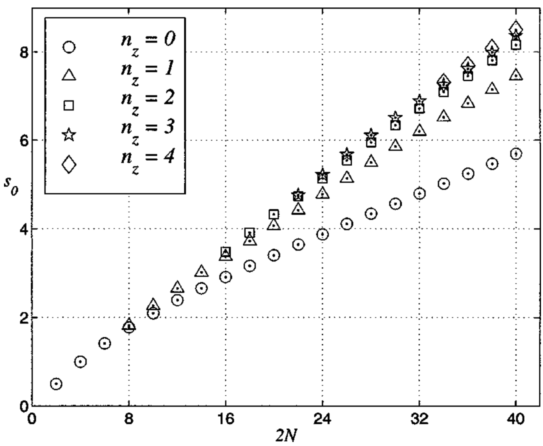

FIG. 1. The Sobolev regularity exponents s_0 from Table I.

As discussed below (see Section 4) the entries in the column $n_z = 1$ are the best possible in the sense that the corresponding wavelets are the smoothest possible among those with $2N \geq 8$ coefficients and at least $N - 2$ vanishing moments.

Extensive numerical tests indicate that for higher values of n_z the exponents listed in Table I are the best possible among those wavelets with $2N$ coefficients and whose scaling filter m_0 has the indicated number of zeros on $(\pi/2, \pi)$. Note also how the results, except for the first few entries, lie on almost straight lines in Fig. 1.

However, it is not completely clear if the results for $n_z \geq 2$ are the best possible if we consider wavelets with the same number of vanishing moments as above but do not insist on m_0 having as many roots on $(\pi/2, \pi)$. We come back to this at the end of Section 4.

We remark that the case $2N = 10$, $n_z = 1$ appears already in [6].

More regular wavelets than the original Daubechies family have been constructed by Volkmer [9] and Lemarié-Rieusset and Zahrouni [5].

Volkmer [9] computes only the asymptotic regularity ratio $\lim_{N\to\infty} \sigma/2N$ and constructs wavelets for which the ratio is close to 0.19. Note that for many entries in Table I the ratio $\sigma/2N$ is over 0.21.

Lemarié-Rieusset and Zahrouni [5] discuss what they call the Matzinger wavelets: the scaling filter has a multiple root at $2\pi/3$ and no other roots on $(\pi/2, \pi)$. Our wavelets are smoother, e.g., the smoothest one they construct has 34 coefficients and Sobolev exponent 6.13. Table I shows that already with $n_z = 1$ our wavelet with the same number of coefficients is more regular. In fact, for each value of N the $n_z = 1$ wavelet listed in Table I is better than the smoothest wavelet discussed in [5].

3. METHODS

We begin by describing the theorem that allows us to compute exactly the Sobolev regularity exponent of solutions to (1). Define another trigonometric polynomial, $r(\xi)$, by factoring out all zeros of $|m_0(\xi)|^2$ at $\xi = \pi$; that is, let

$$|m_0(\xi)|^2 = \left(\frac{1+\cos\xi}{2}\right)^M r(\xi), \tag{4}$$

where $M \in \mathbb{N}$ is chosen so that $r(\pi) \neq 0$. The function r is used to define an operator on $C[0, \pi]$ by

$$T_r u(\xi) = r(\xi/2)u(\xi/2) + r(\pi - \xi/2)u(\pi - \xi/2). \tag{5}$$

The Sobolev regularity exponent of a solution to the dilation equation (1) is given by the following theorem, proved independently by Villemoes and Eirola:

THEOREM 3.1 [4, 8]. *Suppose ϕ is a solution to* (1), *M, r, and T_r are as above, and r satisfies the Cohen criterion. Then $\phi \in H^s$ if and only if $s < s_0 = M - \log_4 \rho(T_r)$, where $\rho(T_r)$ is the spectral radius of T_r on $C[0, \pi]$.*

If r is a polynomial of cosines of degree at most d, i.e., $r(\xi) = \sum_{k=0}^{d} b_k \cos(k\xi)$, then T_r maps the space spanned by $\{\cos(k\xi)\}_{k=0}^{d}$ into itself and the spectral radius of T_r can be calculated from the eigenvalues of a finite matrix [4, 8].

Suppose we want to look at a filter of length $2N$ (i.e., a filter m_0 with at most $2N$ nonzero coefficients) such that m_0 has a zero of order M at π and n_z zeros on the interval $(\pi/2, \pi)$.

Note that M is also the number of vanishing moments the associated wavelet has (if one exists). Define a new sequence of coefficients by

$$|m_0(\xi)|^2 = \sum_{k=0}^{2N-1} a_k \cos(k\xi). \tag{6}$$

The orthonormality condition (3) is equivalent to

$$a_0 = \frac{1}{2}, \qquad a_{2k} = 0, \qquad \text{for } k = 1, \ldots, N-1.$$

The zero of m_0 at π implies (linear) relations among the $\{a_k\}$: there are only $N - M = 2n_z$ independent parameters, denoted by $\mathbf{v} = (v_1, \ldots, v_{n_z})$. Note that there is always exactly one value of $\mathbf{v}$ that corresponds to the Daubechies wavelet (i.e., all roots of m_0 are at π).

4. OPTIMIZATION

With two degrees of freedom optimization was done on the $\mathbf{v} = (v_1, v_2)$ parameters. Since some values of $\mathbf{v}$ yield an $r_{\mathbf{v}}$ that also takes on negative values, this is now a constrained optimization problem with the nonlinear constraint $\min_{0 \le \xi \le \pi} r_{\mathbf{v}}(\xi) \ge 0$.

Numerical optimization of the regularity exponent was done by using the Matlab Optimization Toolbox. The smoothest wavelet was always found on the boundary of the feasible region $\{\mathbf{v} \in \mathbb{R}^2 : \min r_{\mathbf{v}} \ge 0\}$; that is, the optimal r has a double root on $(\pi/2, \pi)$. Since there are only two degrees of freedom, verifying that the optimum is a global one is easy. It is also easy to check that the Cohen criterion is satisfied, e.g., that there are no roots on the interval $(\pi/2, \pi)$.

The set $\{\mathbf{v} \in \mathbb{R}^2 : \min r_{\mathbf{v}} \ge 0\}$ contains a parameterization of all wavelets with $2N$ coefficients and $N-2$ vanishing moments. It is also easy to verify numerically that one degree of freedom does not yield wavelets that are smoother than the Daubechies wavelet with the same number of filter coefficients. Together these results show the regularity exponents in the $n_z = 1$ column of Table I are the best possible among wavelets with at least $N-2$ vanishing moments and $2N \ge 8$ filter coefficients.

With more degrees of freedom the above approach was hard to implement reliably because of the constraint $\min_{0 \le \xi \le \pi} r_{\mathbf{v}}(\xi) \ge 0$ (recall that r is a trigonometric polynomial of high degree). Instead, the optimization was done by considering the Sobolev regularity exponent s_0 as a function of the (double) roots $\mathbf{z} = (z_1, \ldots, z_{n_z})$ of r on $(\pi/2, \pi)$: For a fixed value of $\mathbf{z}$ we can solve for the parameters $\mathbf{v}$ discussed above by requiring $r_{\mathbf{v}}(\mathbf{z}) = r'_{\mathbf{v}}(\mathbf{z}) = 0$. This is a linear system, though it may be badly conditioned when r is of high degree. (This problem was overcome by using Mathematica and 30 digit precision in the computations.)

The benefits are that now we are studying an unconstrained optimization problem and the number of degrees of freedom is halved. Extensive numerical tests show that the results as stated are optimal, that is, in the class of wavelets whose filter has the indicated number of zeros on $(\pi/2, \pi)$. However, since the optimization is done on a subset of the boundary of the feasible region it is not clear if the wavelets listed in Table I with $n_z \ge 2$ have optimal regularity in the larger class of wavelets where the filter has *at most* n_z roots on $(\pi/2, \pi)$.

5. CONCLUSIONS

We have constructed wavelets that are more regular than any of the wavelets that have appeared in the literature with the same number of filter coefficients, thus improving the ratio of regularity to complexity. Our methods show that a numerical optimization approach provides the flexibility necessary to construct highly regular wavelets while keeping the number of vanishing moments of the wavelet as high as possible.

We have also shown our results are optimal for wavelets with $2N$ filter coefficients and at least $N-2$ vanishing moments.

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